

Question_2:

Code:

```
1  /*
2  Question No: 2
3  Given the preorder and inorder traversals of a binary tree,
4  construct the tree and print its postorder traversal.
5  Preorder : Root -> Left -> Right
6  Inorder  : Left -> Root -> Right
7  Postorder: Left -> Right -> Root
8  */
9
10 #include <iostream>
11 using namespace std;
12
13 struct node {
14     int data;
15     node *left,*right;
16 };
17 int pre[100005],in[100005],pos[100005];
18 int p=0;
19
20 node* build(int l,int r) {
21     if(l>r) return NULL;
22
23     int x=pre[p++];
24     node* root=new node{x,NULL,NULL};
25     int m=pos[x];
26     root->left=build(l,m-1);
27     root->right=build(m+1,r);
28     return root;
29 }
30 void postorder(node* root) {
31     if(!root) return;
32
33     postorder(root->left);
34     postorder(root->right);
35     cout<<root->data<<" ";
36 }
37 int main() {
38     int n;
```

```
39     cin>>n;
40
41     for(int i=0;i<n;i++) cin>>pre[i];
42
43     for(int i=0;i<n;i++) {
44         cin>>in[i];
45         pos[in[i]]=i;
46     }
47     node* root=build(0,n-1);
48     postorder(root);
49 }
```

Output:

Test Case_1:

```
5
5 3 2 1 4
3 5 1 2 4
3 1 4 2 5
PS C:\Users\Prabin Yadav\
```

Test Case_2:

```
6
6 5 3 2 1 4
6 3 5 1 2 4
3 1 4 2 5 6
PS C:\Users\Prabin Yadav\Doc
```